

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Gen T Full Connectivity Option
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:
1. Allow optional, full connectivity between Gen T Adapters and USB4 Ports in a Host Router 2. In full connectivity, there is no requirement to have the same number of Gen T Adapters as USB4 Ports 3. Bundle Weight field transitions from Gen T Adapter CS to USB4 Port Capability Fields

Benefits as a result of the changes:
The change allows more flexibility when implementing Gen T Ports and Gen T Adapters.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
More flexibility in implementation choices.

An analysis of the software implications:
Connection Manager should accommodate both connectivity options

An analysis of the compliance testing implications:
None

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Actual Change

(a). 5.2.5 Connectivity Rules

A Router shall be capable of forwarding packets as follows:

- If the *Gen T Full Connectivity Support* bit in the Router Configuration space is set to 0b
 - USB4 Ports and USB3 Gen T Adapters are paired in order of increasing Adapter Numbers. A Router shall be able to forward a packet received on the Lane 0 Adapter of a USB4 Port to the paired USB3 Gen T Adapter. A Router shall be able to forward a packet received on a USB3 Gen T Adapter to the Lane 0 Adapter of the paired USB4 Port.
- Else
 - A Router shall be able to forward a packet received on the Lane 0 Adapter of a USB4 Port to any USB3 Gen T Adapter. A Router shall be able to forward a packet received on a USB3 Gen T Adapter to the Lane 0 Adapter of any USB4 Port.

(b). Table 5-14 Buffer Allocation

Table 5-14. Buffer Allocation Parameters

Name	Description	Egress Adapter	Requirements
baMaxUSB3GenT	The buffer allocation requested, for the sum of all USB3 Gen T Paths through a <u>single USB3 Gen T AdapterUSB4 Port</u> , to achieve the maximum target bandwidth.	USB3 Gen T	Shall be present if Router has a USB3 Gen T Adapter.

(c) 5.3.3.1.1 Path Schedulers

A Path Scheduler in a Host Router that sets the *Gen T Bundle Weight Mode* bit to 1b operates with the following additional rules and changes for USB3 Gen T Paths:

- The shared weight assigned to all the USB3 Gen T Paths shall be determined by the *Bundle Weight* field in the USB4 Port Capability FieldsUSB3 Gen T Adapter Configuration Space of the USB3 Gen T Adapter connected to the USB4 Egress Port.
- When the *Bundle Weight* field changes, the Path Scheduler shall use the new shared weight.
- The Path Scheduler shall support weights in the range of 1-255 for the shared weight of the USB3 Gen T Paths.
- The Path Scheduler shall schedule packets such that the number of bytes scheduled for all the USB3 Gen T Paths together is proportional to the shared weight assigned in the *Bundle Weight* field with 10% maximum deviation over a measurement period of 1 ms.

(d). 4.2.1.6.3 Objections

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A USB4 Port shall assert an objection to enter CL2 state if:

...

- ~~The Lane 0 Adapter is referenced in an Egress Adapter field of a USB3 Gen T Adapter's Routing Table, and all~~ All the USB3 Gen T Ports that are mapped to Paths that goes through the USB3 Gen T Lane 0 Adapter are not in Disabled, U2 or U3 state.
- ~~The Lane 0 Adapter is referenced in an Egress Adapter field of a USB3 Gen T Adapter's Routing Table, one~~ One of the USB3 Gen T Ports that are mapped to Paths that goes through the USB3 Gen T Lane 0 Adapter is in U2 state, and the *U2CL2 Enable* bit in USB3 Gen T Adapter Configuration Space is set to 0b.

...

A USB4 Port shall assert an objection to enter CL1 state if:

...

- ~~The Lane 0 Adapter is referenced in an Egress Adapter field of a USB3 Gen T Adapter's Routing Table and all~~ All the USB3 Gen T Ports that are mapped to Paths that goes through the USB3 Gen T Lane 0 Adapter are not in Disabled, U2 or U3 state.

(e). 4.4.5.2.1 SBRX Goes Low

The Router shall do the following for each enabled Lane Adapter in the disconnected USB4 Port:

...

- ~~If the Lane Adapter is connected to a Downstream USB3 Gen T Adapter, set~~ Set all of the *Path Enable* bits ~~in for~~ the Paths that goes through the disconnected Lane Adapter ~~USB3 Gen T Adapter Configuration Capability of the USB3 Gen T Adapter~~ to their default values.

(f). Table 8-3 Router Configuration Space Basic Attributes

Table 8-3. Router Configuration Space Basic Attributes

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
6	ROUTER_CS_6	23	Reserved <u>Gen T Full Connectivity Support (GTFC)</u> <u>A Host Router shall set this bit to 1b if all of its USB3 Gen T Adapters supports full connectivity as defined in Section 5.2.5. Otherwise, it shall be set to 0b.</u>	RO	<u>0Vendor Defined</u>

**** Editor Note **** Update Figure 8-1

(g). Table 8-13 USB4 Port Capability Fields

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Table 8-13. USB4 Port Capability Fields

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
19	PORT_CS_19	...	...	...	...
		157 :5	Reserved	Rsvd	0
		15:8	Bundle Weight This field is set by the Connection Manager as the total weight for all the enabled Gen T Paths that goes through this USB4 Port. This field only applies to a USB4 Port in a Host Router that has the Gen T Bundle Weight Mode bit set to 1b.	R/W	0

**** Editor Note **** Update Figure 8-10

(h). Table 8-15 USB3 Gen T Adapter Configuration Capability Fields

Table 8-15. USB3 Gen T Adapter Configuration Capability Fields

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
1	ADP_USB3_GT_CS_1	7:0	Bundle Weight This field is set by the Connection Manager as the total weight for all the enabled Gen T Paths that originate from this Gen T Adapter. This field only applies to a USB3 Gen T Adapter in a Host Router that has the Gen T Bundle Weight Mode bit set to 1b.	R/W	0
		31:80	Reserved	Rsvd	0

**** Editor Note **** Update Figure 8-12

(i). 9 USB3 Tunneling

Gen T System Level Requirements

USB4 Hosts:

A USB4 Host may optionally support USB3 Gen T tunneling. The Host Router in a USB4 Host that supports Gen T tunneling shall have at least one Downstream USB3 Gen T Adapter. In case the Host Router does not support full Gen T connectivity (Gen T Full Connectivity Support bit is set to 0b), it shall have one Downstream USB3 Gen T Adapter per USB4 Downstream Facing Port. Each Downstream USB3 Gen T Adapter shall interface to at least one downstream USB3 Gen T Port on the host controller.